MOHAMMAD KAMRUL HASAN

Systems & ML Infrastructure Engineer • Backend • On-Device AI

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Summary

CSE undergraduate at BUET (CGPA 3.77) working toward his own LLM inference engine. Made llama.cpp decode **2.2–2.6× faster** with up to **3.3× less energy** on Android phones, and benchmarked retrieval models for production at an AI startup. Backend engineer across NestJS, FastAPI, Spring Boot, PostgreSQL, MongoDB, Redis and RabbitMQ. Co-author, UbiComp/ISWC 2026.

Education

Bangladesh University of Engineering and Technology (BUET) **April 2025 – Present**
B.Sc. in Computer Science and Engineering *Dhaka, Bangladesh*
CGPA: **3.77**/4.00 • Ranked **79th** nationwide in the BUET admission merit list

Experience

Backend Developer Intern **June 2026 – Present**
Version3.ai • *Conversational-commerce AI for online stores* *Dhaka, Bangladesh*

- Own the **multi-tenant database platform** end to end, keeping every store's data strictly isolated, using **NestJS**, **MongoDB** and **RabbitMQ**.
- Built the **WooCommerce** and **Shopify** integrations, so stores connect securely in one click (OAuth PKCE, signed requests, encrypted secrets, a retry-safe job queue).
- Fixed a bug that made WooCommerce **permanently delete** a store's API keys on connect, caused by the server returning the wrong HTTP status.
- Picked the image model for visual product search by benchmarking 4 models on **132K product images**; the winner finds the right product first try **67%** of the time.
- Showed vector compression in **Qdrant** costs almost no accuracy (median −0.4 pts), so search can run on cheap CPU servers.

Software Engineering Intern **Dec 2025 – Mar 2026**
Edulytics *Dhaka, Bangladesh*

- Shipped several core user flows for a **React Native** MVP on tight sprints, ramping up quickly on a large existing codebase.
- Worked through feature branches and pull requests with a remote team.

Publications

From Wearable Data to Risk Literacy: Visualizing Physiological Heat Stress for Rickshaw Pullers in Dhaka
M. Rahaman, S. S. Rahman, M. S. M. Noor, K. S. S. Siddiquee, **M. K. Hasan**, et al. **UbiComp/ISWC 2026**
ISWC Notes & Briefs, ACM • doi.org/10.1145/3830727.3834827 • ORCID

- Turned 3 years of wearable data from 100 rickshaw pullers into heat-risk visuals that low-literacy workers can read; I transcribed the interviews behind the qualitative analysis.

Projects — AI / ML Systems

LITTLETai — Faster, Lower-Power LLM Inference on Android Phones *2026*
[GitHub • Arm Create: AI Optimization Challenge] *Shell* • *Python* • *llama.cpp* • *KleidiAI* • *simpleperf* • *perf*

- Made llama.cpp decode **2.2–2.6× faster** with **2.6–3.3× less energy** (8.5–11× on Arm's KleidiAI build) by picking the right CPU cores and thread counts for phones with mixed big and small cores.
- Found why the default was slow: with all 8 cores busy, threads burned up to **5× more CPU instructions** spin-waiting on each other; also found llama.cpp's core-pinning flag silently did nothing on Android.
- Built a repeatable benchmark rig (thermal cooldowns, randomized runs, energy and CPU-counter logging, 209 tests) so every number can be regenerated exactly.

NodeCommerce — Demand Forecasting for Bangladesh *2026*
[GitHub • AI Buildfest hackathon] *Python* • *FastAPI* • *LightGBM* • *SHAP* • *Neo4j* • *PostgreSQL* • *Redis* • *Docker*

- Forecast stock demand for **495 upazilas** across all 64 districts with LightGBM on 94 features, with SHAP showing *why* each forecast moved.
- Automated 8 daily-to-weekly pipeline jobs (features, inference, spike detection, retraining) and a map dashboard of regional surplus and shortage.

DroneCam — Spotting Tiny People and Cars in Drone Footage *2026*
[GitHub] *Python* • *YOLO* • *SAHI* • *ByteTrack* • *BoT-SORT*

- Reached **0.728 mAP@0.5** on VisDrone by training at high resolution and slicing frames into tiles, so distant objects aren't lost when images shrink.
- Offered two tracking modes: an accurate one for analysis and a fast one for real time.

Sentinel — Offline Medical Triage App *2025*
[GitHub • Lead developer, team of 3] *React Native* • *TypeScript* • *TensorFlow Lite* • *MiniLM*

- Grades symptoms as Critical, Urgent or Routine **with no internet**, using a small CNN plus a **20 MB** int8 language-model fallback when the CNN is unsure.

Bayyinah-AI — Semantic Search API *2025*
[GitHub] *Python* • *FastAPI* • *sentence-transformers* • *FAISS* • *Hugging Face Spaces*

- Matches free-text queries to relevant Quranic verses in **∼150 ms** using vector search plus re-ranking for relevance and variety, with backup LLM providers so it never goes down.

Poramorshok — AI Health Companion *2026*
[GitHub • Hackathon] *React* • *TypeScript* • *Supabase*

- Built a privacy-first health app (symptom checker, diet and exercise plans, reminders) in English and Bengali, with tested Supabase auth and sync.

Projects — Backend, Databases & Systems

Ilham — Hadith Study Platform *2026*

[GitHub • DBMS term project] *PostgreSQL • PL/pgSQL • Node.js • Express • React*

- Loaded **14,941 hadiths** and **139,766** narrator-chain links with **99.6%** of narrators matched, using a custom ETL into a PostgreSQL database.
- Scored each hadith's reliability in SQL from its weakest narrator, and kept the reference data read-only so the app can never corrupt it.

Bayyinah — Collaborative Desktop Quran App *2025*

[GitHub] *Java • JavaFX • Spring Boot • WebRTC • SQLite • PostgreSQL*

- Instant full-text search over all **6,236 verses** offline, plus live group reading sessions with real-time audio.

BondhuPath — Real-Time Community Backend *2026*

Bun • ElysiaJS • PostgreSQL • Drizzle • WebSockets

- Built login that can revoke any session instantly and refresh tokens without dropping live connections, plus online-presence tracking per room.

AyahBot v4 — Bot Platform *2025*

TypeScript • Rust • gRPC • PostgreSQL

- Split the bot into a TypeScript gateway and Rust services, with a PostgreSQL job queue so many workers can run without clashing.

KXCache & Kafka Clone — Systems From Scratch *2026*

C++ • Java

- Building a Redis-style cache server in C++ and a Kafka-style message broker in Java, to learn non-blocking I/O and distributed messaging hands-on.

Q*bert — 3D Arcade Game *2025*

[GitHub • Team of 2, BUET] *C++ • OpenGL*

- Wrote the 3D graphics engine and enemy AI for a full remake of the arcade game, and led the project.

Embedded & Reverse Engineering *2025 – 2026*

ESP32-S3 • ESP-IDF • x86 Assembly • Ghidra

- Writing my own OLED display driver for ESP32-S3; analyze x86 binaries and play pwn/RE CTFs.

Technical Skills

Languages: C, C++, Python, Java, TypeScript, JavaScript, SQL, Rust (working), Shell, x86 Assembly
AI / ML: llama.cpp, quantization (GGUF, int8), TensorFlow Lite, PyTorch, Hugging Face, QLoRA, SigLIP / CLIP, FAISS, Qdrant, LightGBM, YOLO, OpenCV, LangGraph
Performance: perf, simpleperf, Perfetto, Arm Streamline, benchmark design
Backend: NestJS, FastAPI, Spring Boot, Express, ElysiaJS, gRPC, REST, WebSockets, WebRTC, OAuth, JWT
Data & Infra: PostgreSQL, MongoDB, Redis, RabbitMQ, Neo4j, SQLite, Supabase, Docker, GitHub Actions, Linux
Frontend & Mobile: React, React Native, Next.js, JavaFX, OpenGL